

Aditya Sathishkumar

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EDUCATION

University of Massachusetts Amherst

Amherst, MA

Bachelor of Science in Computer Science | **GPA: 3.84**

Expected May 2027

Coursework – Data Structures, Algorithms, Software Engineering, Object-Oriented Programming, Computer Systems, Stats

Extracurricular – ML Club (Robotics Lead) on multi-agent ML, UMass Robotics (developer) on NASA Lunabotics rover

EXPERIENCE

Software Engineer Intern

Jun 2026 – Aug 2026

Blue Origin

Kent, WA

- Architected real-time performance monitoring system for hardware-in-the-loop (HIL) lab in **C++** and **Python**, automating HIL investigation with **anomaly detection**, notifications, and AI-assisted **RCA** via telemetry insight on **60+** hosts
- Engineered **event-driven** metrics pipeline using **RabbitMQ**, async HTTP, **UDP**, and **multithreading**, capturing metrics at **250 ns** overhead out of **3.33 ms** event loop budget and owning design through iterative delivery
- Crafted **fault-tolerant** infrastructure with automated retries, backoff, error isolation, thread-safe caching, and CPU core partitioning, preventing monitoring failures from disrupting critical HIL execution
- Modularized** metrics capture via **publish-subscribe** adapters, decoupling **Datadog** to eliminate vendor-lock and enable hot-swappable backends, deploying via **GitLab**, **CI/CD**, **Artifactory**, **Docker**, and **Ansible** to fit existing workflow
- Validated system resilience via **Pytest** and **GTest**, executing **failure injection** scenarios in a virtual HIL environment to guarantee strict error containment and zero-propagation behavior

NASA

Aug 2025 – Dec 2025

Software Engineer Co-op

Merritt Island, FL

- Enhanced **Python** V&V framework in Jama for Artemis, automating traceability for **10,000+** flight-critical requirements
- Used **REST APIs** synchronizing flight software implementations with SQL databases for automated requirements rollup
- Mitigated **40-day** federal shutdown schedule compression by prioritizing core infrastructure over secondary scope
- Built **Linux** data pipeline transforming raw **XML** test histories into structured datasets for audits and anomaly analysis

Software Engineer Co-op

Jan 2025 – Jul 2025

Huntington Ingalls Industries (HII)

Dayton, OH

- Engineered **C++** aircraft tracker and ADS-B receiver, capturing truth data to train **YOLO CNNs** for aerial refueling
- Integrated **thread pools** and downsampling to process **5 Gbps** camera feeds, optimizing gimbal tracking loop with **Boost**
- Architected **C** RF receiver using **UDP** streamer patterns and callbacks to process packets instantly, reducing latency **35%**
- Programmed **C++ API** with **std::chrono** microsecond precision to resolve **ADS-B** signal collisions, reducing data overlap **20%**; validated tracking reliability through rigorous unit, simulation, memory, and live hardware field tests
- Developed real-time **C++ engine** simulating kinematic flight tracks for custom playback and **CNN** output validation, streamlining the data pipeline and reducing model iteration time by **80%**

PROJECTS

Swarm Hunt | *Python, Mutli-agent, Machine Learning, Simulation, PettingZoo, AgileRL*

May 2026

- Engineered continuous-control physics engine for **6-agent** pursuit, processing **kinematics** and collisions at **60 Hz**
- Formulated dense **reward functions**, training emergent swarm tactics via **PettingZoo** and **AgileRL**
- Architected a **ParallelEnv** integration with vectorized **spatial observations**, validating API logic via **Pytest**

Reactive Boids Engine | *Python, Spatial Partitioning, Physics Simulation*

Jun 2024

- Engineered custom physics engine at 60 FPS using vector alignment and cohesion formulas to simulate **500+ agents**
- Optimized collision detection via **spatial grid partitioning**, reducing time complexity from **$O(N^2)$** to **$O(N)$**
- Decoupled **behavioral logic** from rendering, enabling a headless execution framework for parallel **ML** pipelines

UMass Robotics | *NASA Lunabotics*

Sep 2023

- Designed dual-camera computer vision system for **ROS** rover, configuring orthogonal cameras to reduce depth noise
- Outlined stereo visual telemetry with onboard compute to enhance autonomous navigation reliability across lunar terrain

SKILLS

Languages: C++, C, Python, SQL, Bash, XML, YAML

Frameworks & Libraries: Boost, OpenCV, FastAPI, Celery, aiohttp, PettingZoo, AgileRL, CNNs, Pydantic, Pytest, GTest

Systems & Architecture: Hardware-in-the-Loop (HIL), RTOS, Software-Defined Radio (SDR), RabbitMQ, Datadog, REST APIs, UDP, Publish-Subscribe, Dependency Injection, Multithreading, Thread Pools, Fault Tolerance

Tools & Infrastructure: Linux, Docker, Ansible, GitLab CI/CD, Artifactory, Jama, CouchDB, Traefik, MCP Servers